Final Recommendations

1. Machine Health & Maintenance Recommendations

Hydraulic Pressure Monitoring

All analyses (boxplots, correlations, feature importance, and interactions) show that hydraulic pressure is the strongest indicator of downtime.
Actions:

- Perform weekly hydraulic system inspection.
- Check pumps, valves, and potential oil leaks.
- Add a low-pressure alert sensor.
- Recalibrate the hydraulic system if pressure drops below 90 bar.

Reduce Cutting Force on High-Risk Days

Cutting force increases right before failures, forming a strong positive correlation.
Actions:

- Reduce cutting speed when pressure is low.
- Inspect the cutting tool condition.
- Avoid heavy machining operations during high-risk periods.

Monitor Torque and Load Index Variability

Torque and load index shrink to lower ranges and show abnormal variability on failure days.
Actions:

- Adjust spindle speed and review spindle alignment.
- Inspect spindle bearings if vibration + low torque appear.
- Track daily load index variability.

Handle Vibration Spikes (Especially May–June)

Vibration shows heavy fluctuation during May–June, indicating mechanical instability.
Actions:

- Schedule machine recalibration before summer.
- Inspect cooling system and spindle bearings.
- Set vibration threshold alerts.

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2. Predictive Maintenance Recommendations (Per Machine)

Machine L1 – High Risk

- **Downtime probability: 79% (High)**
- Shows strong degradation + vibration spikes.

Actions:

- Perform maintenance within 24 hours.
- Reduce batch production for the next day.
- Avoid heavy-load machining.
- Check hydraulic pressure and cutting force sensors.

Machine L2 – Low Risk

- **Downtime probability: 10% (Low)**
- Stable trends.

Actions:

- Continue normal production.
- Standard monitoring only.

Machine L3 – Moderate Risk

- **Downtime probability: 25% (Moderate)**
- Slight degradation in torque and pressure ratios.

Actions:

- Monitor pressure closely.
- Avoid high cutting force operations.
- Keep batch count normal but monitor behavior.

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3. Operational Recommendations

Batch Scheduling Optimization

- High-risk day → **reduce production by 30–40%**
- Medium risk → operate at low load
- Low risk → normal operation

Daily Early Warning Indicators

Trigger a "Warning Mode" if 2 of these conditions occur:

- Hydraulic Pressure < 90
- Cutting Force > 2.8 kN
- Torque < 22 Nm

Cooling System Inspection

Interaction analysis shows:
High coolant temperature + low pressure = 85% downtime probability

Actions:

- Clean cooling system every 2 weeks
- Add coolant temperature alerts
- Check coolant flow rate

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0 4. Strategic & Long-Term Recommendations

Deploy a Real-Time Monitoring Dashboard

Should include:

- Pressure
- Torque
- Load
- Forecast probability
- Batch recommendations
- Failure alerts

Preventive Maintenance Schedule

Based on behavior:

- L1 → every 10–14 days
- L2 → every 20–25 days
- L3 → every 15–18 days

Invest in Additional Sensors

- Hydraulic pressure sensors
- Vibration sensors
- Temperature sensors

Will significantly improve forecasting accuracy.